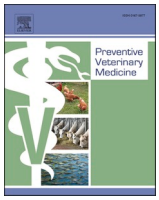
Preventive Veterinary Medicine 220 (2023) 106049



Contents lists available at ScienceDirect

Preventive Veterinary Medicine

journal homepage: www.elsevier.com/locate/prevetmed





Large scale spatio-temporal modelling of risk factors associated with tuberculosis exposure at the wildlife-livestock interface

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ARTICLE INFO

Keywords:
Bovine tuberculosis
Epidemiology
Risk factors
Regionalization
Variation partitioning
Wildlife-livestock interface

ABSTRACT

The management of animal tuberculosis (TB) is a priority for European Union animal health authorities. However, and despite all the efforts made to date, a significant part of Spain has as yet been unable to obtain the officially tuberculosis-free (OTF) status. Information regarding wildlife disease status is usually scarce, signifying that the role played by wildlife is usually ignored or poorly assessed in large-scale TB risk factor studies. The National Wildlife Health Surveillance Plan in Spain now provides information on infection rates in wildlife reservoirs at a national level, but there are limitations as regards the sample size, the spatio-temporal distribution of the samples, and the lack of homogeneity of the diagnostic techniques employed. The objective of the study described herein was, therefore, to employ a Bayesian approach with the intention of identifying the risk factors associated with four TB rates in cattle: prevalence, incidence, maintenance and persistence in Spain during the period 2014–2019. The modeling approach included highly informative spatio-temporal latent effects with which to control the limitations of the data. Variation partitioning procedures were carried out, and the pure effect of each factor was mapped in order to identify the most relevant factors associated with TB dynamics in cattle in each region. This made it possible to disclose that the movement of cattle, particularly from counties with herd incidence $> 1\%$, was the main driver of the TB dynamics in cattle. The abundance of herds bred for bullfighting was retained in all four models, but had less weight than the movements. After accounting for farm-related factors, the TB prevalence in wild boar was retained in all the models and was significantly related to incidence, maintenance and persistence. With regard to the incidence, variation partitioning revealed that wildlife was the most explicative factor, thus suggesting that it plays a role in the introduction of the pathogen into uninfected herds, and consequently highlighting its importance in breakdowns. These results show, for the first time on a national scale, that wild ungulates play a relevant role in the spatio-temporal variability of TB in cattle, particularly as regards their disease status. Moreover, the spatial representation of the pure effect of each factor made it possible to identify which factors are driving the disease dynamics in each region, thus showing that it is a valuable tool with which to focus efforts towards achieving the OTF status.

1. Introduction

Animal tuberculosis (TB) is a chronic infection that is caused by *Mycobacterium bovis* and related members of the *Mycobacterium tuberculosis* complex (MTC), and can infect a wide range of domestic and wild mammals and humans (Rodriguez-Campos et al., 2014). The zoonotic potential of TB and its economic importance owing to its impact on the livestock trade signify that the European Union has spent many decades attempting to control it (Allen et al., 2018). Most European countries have now obtained the officially TB free status as a result of the measures implemented (OTF; see EFSA, 2022). However, despite the decline in prevalence, some regions have experienced difficulties in attaining the OTF status, and the disease is still prevalent in Eastern Europe, the British Islands, Italy, Greece, and the Iberian Peninsula. The difficulties involved in managing the disease are related principally to the testing procedure employed, cattle management, wildlife reservoirs and the intrinsic characteristics of the MTC, with differences in their relevance among regions (Guta et al., 2014; Allen et al., 2018).

The role played by wildlife as regards hindering the eradication of TB in cattle has been widely studied (Palmer et al., 2012; Gortazar et al., 2014; Allen et al., 2018). Several wild species can act as MTC reservoirs, such as the Eurasian wild boar (*Sus scrofa*) and the red deer (*Cervus*

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https://doi.org/10.1016/j.prevetmed.2023.106049
Received 2 March 2023; Received in revised form 4 September 2023; Accepted 14 October 2023
Available online 17 October 2023
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elaphus) in the Iberian Peninsula, the European badger (*Meles meles*) in the United Kingdom and the Republic of Ireland, the white-tailed deer in North America (*Odocoileus virginianus*) and the brushtail possum (*Trichosurus vulpecula*) in New Zealand (Gortazar et al., 2012; Palmer et al., 2012). Measures targeting wildlife reservoirs have entailed reductions in the risk of cattle herds being exposed to TB (Livingstone et al., 2015; Allen et al., 2018), and wildlife should, therefore, be considered in livestock epidemiological studies, thus making disease monitoring in wildlife an essential part of the integrated control of TB (McDonald et al., 2008; Boadella et al., 2011b). However, data series regarding disease status in wildlife are scarce worldwide, and TB is no exception (Gortazar et al., 2014). The role played by wildlife in TB maintenance at the wildlife-livestock interface has, therefore, received attention at mostly a regional and local extent (Walter et al., 2014; Glaser et al., 2016; Gortazar et al., 2017; Salvador et al., 2019). However, the factors driving the TB dynamics in each region could be different from those that act on greater scales, and the implication of wildlife in disease dynamics should, therefore, be studied on various scales (Garabed et al., 2019; Tao et al., 2021). In this respect, serological surveys of TB in wild animals have been carried out with national scope (Boadella et al., 2011a; Miller and Sweeney, 2013; Richomme et al., 2013), and a proxy of the tuberculosis hazard that wildlife poses to cattle has been evaluated at the European level (Hardstaff et al., 2014). However, the number of samples and the study design in these studies have normally prevented the spatial and temporal dynamics from being assessed conjointly. In addition, information regarding wildlife disease status has not previously been implemented in risk factor assessments of TB in cattle by considering the spatial and temporal scales. All of the above has limited the large-scale study of the role played by wild animals in TB dynamics in cattle to the inclusion of information regarding their presence and/or abundance (Garcia-Saenz et al., 2014; Wright et al., 2015; Madden et al., 2021). The lack of the continuous and systematic concurrent large-scale surveillance and monitoring of shared diseases in where both livestock and wildlife are present hampers the understanding of their respective roles in the spatio-temporal epidemiology of TB, which is essential if targeted recommendations with which to achieve OTF status are to be provided (Cardoso et al., 2021).

TB prevalence in cattle has stabilized in Spain in recent years, despite the efforts made to control the disease (Anon, 2021). The National Wildlife Health Surveillance Plan (NWHSP), established in 2011, and the Action Plan on Tuberculosis on Wild Species (PATUBES), established in 2017 (Real, 2020), were developed in an attempt to improve the knowledge regarding the status of the disease in wildlife and mitigate its effects, thus providing information on infection rates in wildlife on a national scale.

Considering the potential role of wildlife in disease dynamics, to implement the novel information available regarding the health status of wild reservoirs at a national scale in TB risk assessments for livestock could offer valuable insights into enhancing our understanding of the spatio-temporal dynamics of the disease. In this context, our objectives were to: i) identify the risk factors associated with four TB status–related response variables for cattle at a national level in Spain: prevalence, incidence, maintenance, and persistence, ascertaining whether wildlife is playing a role in large-scale disease dynamics, and ii) regionalize the weights of the risk factors in order to provide a basis for further disease monitoring and management efforts for achieving the OTF status. Information regarding the disease status in wildlife on a national scale provided by the NWHSP was, therefore, included for the first time. However, the challenges involved in wildlife disease sampling (Boadella et al., 2011b; Jamin and Riviere, 2020) resulted in several limitations as regards conducting analysis using the NWHSP data, among which stand out the sample size, sometimes reduced or even non-existent, the irregular spatial and temporal distribution of sample collection, and the lack of homogeneity in diagnostic methods. Considering that the NWHSP program data was being assessed for the first time, we also aimed to iii) develop a statistical approach with which to deal with the common limitations of wildlife disease data, and iv) analyze the temporal trend of the TB status in wild boar.

2. Materials and methods

2.1. Definition of response variables

Cattle testing is part of the National Program for the Bovine Tuberculosis Eradication. It is performed principally by means of a single intradermal tuberculin test (IDT), followed by confirmation using an interferon-γ assay (IFN-γ) when considered necessary (Anon, 2013). Cattle of over 6 months of age are tested from three times a year to once every two years, depending on the prevalence in the herd and the prevalence in their region (see Table S.1 in Appendix 1 in Supporting Information). All animals that attain a positive result to the test are culled (Anon, 2013, 2021). The Spanish Ministry of Agriculture, Food and Fisheries (MAPA) has provided data for the period 2014–2019 regarding the number of herds tested and the number of herds that tested positive and new positives each year, in which new positives are considered to be those herds that tested positive after 1 or more years testing negative.

Four TB-status parameters on cattle farms were studied in order to attain a further understanding of the relationship between the predictors and the disease dynamics. These were: (1) prevalence, defined as herds that tested positive divided by the number of herds tested; (2) incidence, defined as new positive herds divided by the number of herds that tested negative the previous year; (3) maintenance, defined as whether the disease is present in the county (at least one herd has tested positive) consecutively to the previous year, and (4) persistence, defined as the number of consecutive years that a given county had positive herds in the period studied (2014 – 2019). The first three parameters represent a value per year and county, while the last one is a total count value per county.

2.2. Selection of predictors

TB in cattle has previously been related to cattle abundance, cattle movements, and wildlife (Humblet et al., 2009; Bessell et al., 2012; Palmer et al., 2012; Skuce et al., 2012; Broughan et al., 2016). It has been observed that the cattle abundance effect may differ among production types, particularly in the case of cattle used for bullfights (Skuce et al., 2012; Martinez-Lopez et al., 2014; Broughan et al., 2016). Cattle abundances by production type (beef, dairy and bullfight), as provided by the MAPA, were consequently included as predictors.

Cattle movements can pose a greater risk when they originate from areas with a high disease incidence rate (Gates et al., 2013). We therefore divided cattle movements according to the county of origin: movements originating from counties with a herd incidence of $> 1\%$ were considered high-risk (HR), while all others were considered low-risk (LR; Garcia-Saenz et al., 2014). Considering that most movements occur between farms within the same county, we also considered external movements as potential predictors, which are those in which the origin and destination counties are different. These external movements were also divided into high-risk (EHR) and low-risk (ELR) following the same criterion.

With regard to wildlife, both abundance and disease status have been shown to be associated with TB in cattle (Rodriguez-Prieto et al., 2012; Martinez-Lopez et al., 2014; Walter et al., 2014; Ramsey et al., 2016). We consequently included the wild boar hunting yield (ENETWILD--consortium et al., 2020) and red deer environmental favorability (Acevedo et al., 2010) as proxies of relative abundance of these two species (Imperio et al., 2010), which are considered the main wild reservoirs of TB in Spain (Gortazar et al., 2012). TB prevalence in wild boar was also included, estimated as the proportion of wild boars that tested positive in the NWHSP with respect to the total number of wild boar tested. Although it was not possible to include TB prevalence in red deer given

the sample size and its spatio-temporal distribution, TB prevalence in wild boar and in red deer are highly correlated in Spain (Vicente et al., 2007).

The wild boar hunting yield was obtained as a 10 × 10 km grid and was aggregated at county level by means of the average value of the grid cells intersecting the county. Red deer habitat favorability was obtained at municipality level and aggregated at county level by employing the mean values of the municipalities intersected by the county weighted by the area intersected.

All fixed effects were standardized by subtracting the variable mean from each value and dividing the result by the standard deviation of the variable. The fixed effects considered for modeling are detailed in Table 1.

To account for other components that may be related to the variability of disease risk, we also included four highly informative spatial, temporal and spatio-temporal latent effects: (1) structured spatial (spatial autocorrelation), for area-associated risks, (2) unstructured spatial (county), for management-associated risks, (3) temporal (year) for risks associated with temporal changes at the national level, and (4) space-time interaction, for specific features in a given county and year (Lawson et al., 2003). Only the spatial effects were considered in the persistence model, since the sample unit in the model is county, lacking temporal variation.

The structured spatial effect incorporates in the model to what extent the value of the response variable at a given location can be explained by the values it takes in the surrounding locations. It was defined by means of Stochastic Partial Differential Equations (SPDE) using Delaunay triangulation (See Appendix 2, Lindgren et al., 2011). Temporal and space-time interaction latent effects were defined by employing a first order autoregressive integrated moving average model (ARIMA), capturing the explanatory capacity of the previous year's value. Heterogeneity (unstructured spatial latent effect) was defined as an independent and identically distributed Gaussian random effect (iid), considering the explanation provided by the average value of the location throughout the study period.

2.3. Dealing with wildlife data limitations

The NWHSP includes information on wild animals sampled and tested for TB between 2014 and 2019: year, county, species, tissue, diagnostic test employed and results, providing 72,860 records of TB tested individuals that were summarized by county and year for the analysis. However, the counties sampled were not consistent across years, resulting in only a few counties with complete time series data. In some cases, the number of animals tested annually per county was lower than 10 individuals. The main wildlife species sampled was wild boar (n = 57308), which were tested mainly by employing ELISA (n = 19849; Aurtenetxe et al., 2008), bacteriological cultures (n = 17340; Romero et al., 2008), macroscopic inspections (n = 9453; Martin-Hernando et al., 2007) and PCRs (n = 7448; Wilton and Cousins, 1992). Although the ELISA employed detect antibodies and not antigens, the chronic nature of TB and the sensitivity and specificity of the test (from 80% to 90% and from 96% to 100% respectively, depending on the test used) has made it a reliable method for diagnosing TB in wild boar (Richomme et al., 2019; Thomas et al., 2021).

The NWHSP shows three main limitations: the sample size, the spatio-temporal distribution of the samples, and the heterogeneity of the diagnostic techniques. In order to reduce bias derived from a low sample size, NWHSP county-year records in which fewer than 10 wild boar were tested were excluded from the modeling (n = 906). This minimum sample size was calculated employing the Epitools proportion calculator (Sergeant, 2018) as that required to estimate prevalence in large populations, with 6% of TB seroprevalence expected for wild boar in most regions in Spain (Boadella et al., 2011a; Gortazar et al., 2017), with a confidence interval of 95% and an accepted error of 15%. Although a higher minimum sample size would have been desirable for wild boar, in this case it would have led to a highly reduced dataset for the modeling, thus diminishing its representativeness. Using the selected minimum sample size, 66% of the remaining records (n = 598) would have had less than 10% error, and 20% (n = 185) would have had less than 5% error.

Prior to modeling, the representativeness of the 906 records selected for modeling with respect to the whole dataset was verified by comparing, year by year, the mean TB prevalence and incidence in cattle in the counties in both datasets –those used for modeling and overall- through the use of an unpaired two-sample t-test. The potential impact of the spatio-temporal variability in the diagnostic techniques should be gathered by the latent effects, so we hypothesized that this would not introduce bias in the results. To confirm this assumption, we checked for differences in the residuals associated to the diagnostic technique using a one-way ANOVA test.

Since it was the first time that this dataset was being analyzed, we deemed it appropriate to assess the temporal trend of TB prevalence in wild boar. However, the spatial variation of the sampling locations throughout the study period makes it challenging to compare the prevalence among years. For this reason, it was assessed by means of statistical modeling, assuming a binomial distribution in the response variable, and considering county as a spatial latent effect in order to reduce bias as a consequence of the sampling design. Two models were performed: (1) year as a categorical predictor, oriented toward discovering the variations between years, and (2) year as a continuous predictor, in order to identify general decreasing or increasing trend

Table 1
Predictors considered as potential drivers of the TB dynamics in cattle and factors into which they are grouped for the variation partitioning procedures.

Factor	Fixed effects	Description	Resolution	Source
Cattle abundance	Beef herds	Number of beef herds	County - year	MAPA
	Dairy herds	Number of dairy herds	County - year	MAPA
	Bullfight herds	Number of bullfight herds	County - year	MAPA
Cattle movements	HR movements	Overall number of movements received on county farms (including movements within the county) originating from counties with incidence > 1%	County - year	MAPA
	LR movements	Overall number of movements received on county farms (including movements within the county) originating from counties with incidence ≤ 1%	County - year	MAPA
	EHR movements	Number of movements received on county farms originating from other counties with incidence > 1%	County - year	MAPA
	ELR movements	Number of movements received on county farms originating from other counties with incidence ≤ 1%	County - year	MAPA
Wild ungulates	Wild boar ab.	Wild boar hunting yield projection	10x10km grid	ENETWILD-consortium et al. (2020)
	Red deer ab.	Red deer habitat favorability projection	Municipality	Acevedo et al. (2010)
	TB in wild boar	TB prevalence in wild boar	County-year	NWHSP (MAPA)

patterns throughout the studied period.

2.4. Risk factor model specifications

With regard to the modeling of risk factors for TB in cattle, where *i* is the county and *j* is the year, the number of herds tested positive (Pr_{ij}) and new positive (I_{ij}) were assumed to follow a binomial distribution with n_{ij} tested herds and m_{ij} tested herds after testing negative for 1 or more years, with a π_{ij} probability of being positive or a γ_{ij} probability of being a new positive. Maintenance (M_{ij}) was assumed to follow a Bernoulli distribution with a φ_{ij} probability in a given county and year of having at least one positive-tested herd after this also having occurred the previous year. Persistence (Ps_i) was assumed to follow a binomial distribution with a ρ_i probability for a given county of having positive-tested herds in consecutive years, and 5 years as the maximum number of trials, given the 6-year study period.

$Pr_{ij} \sim \text{Binomial}\ (n_{ij},\ \pi_{ij})$

$I_{ij} \sim \text{Binomial}\ (m_{ij},\ \gamma_{ij})$

$M_{ij} \sim \text{Binomial}\ (1,\ \varphi_{ij}) = \text{Bernoulli}\ (\varphi_{ij})$

$Ps_i \sim \text{Binomial}\ (5,\ \rho_i)$

where the probability is defined as:

$$\text{logit}\ (\pi_{ij} \text{ or } \gamma_{ij} \text{ or } \varphi_{ij}) = \beta_0 + \beta X + \beta Y_i + \beta Z_j$$

$$\text{logit}\ (\rho_i) = \beta_0 + \beta X + \beta Y_i$$

where, in each model, β_0 is the intercept, βX is the matrix of fixed effects, βY_i is the matrix of spatial latent effects and βZj is the matrix of temporal latent effects.

The final models were obtained by first carrying out a step-forward procedure for fixed effect selection on the basis of the Deviance Information Criterion (DIC; Spiegelhalter et al., 2002). When at some step a model with a DIC value of at least five units lower than the previous one was not obtained, the step-forward was stopped and the model with the lowest DIC was selected (Donovan and Mickey, 2019). Rather than excluding correlated predictors before the modeling, when a new predictor was included in each step, highly correlated predictors (correlation coefficient $>|0.6|$) were excluded from the following steps (See Appendix 3).

Once the fixed effects had been selected, the same procedure was carried out for the latent effects. Considering that high precision in the latent effects results in low influence of those included in the model, we employed a shrinkage prior by specifying a log Gamma (0.1, 0.01) prior distribution for the precision, with the aim of shrinking small effects towards zero.

For the fixed effects selected, the posterior probability distributions of the odds ratio were calculated. A relation was considered significant when the 95% credible intervals (CR) of the odds ratio of posterior probability distribution did not include 1. Values above 1 imply that the predictor increases the risk of infection, new infection, maintenance, or persistence of the disease, while values below 0 imply that the risk decreases. The marginal variance (σ^2) of the latent effects was obtained from the inverse of the posterior distributions of the precision (τ). In the case of the SPDE, the marginal variance was obtained using the "inla.spde2.results" function from the R-INLA package (Rue et al., 2009).

2.5. Variation partitioning

Knowing the weight of each risk factor on the spatiotemporal sample units helps focus efforts on disease management. For this purpose, the fixed effects were grouped into three factors: cattle movements, cattle abundance, and wild ungulates (See Table 1). The spatial latent effects were grouped into a fourth factor. Partial models were carried out for all of the four models finally selected in the step-forward procedure, and all the factors included in the models and the possible combinations among them were analyzed by following the rules of the variation partitioning procedure (Hortal et al., 2008). Briefly, the R^2 obtained from the model represent the variation explained by the factor or combination of factors included in the partial model. The percentage of variation explained by each factor, relative to the total variation explained by the overall factors, was calculated by following the procedure described in Fig. 1 with the objective of comparing the variation explained by the factors. The explanatory contribution of each factor can be decomposed into: (1) pure effect, which is the proportion of the variation that can be explained only by the inclusion of the given factor, and (2) overlaid effect, which is the proportion of the variation explained by the combination of two or more factors (Heikkinen et al., 2005; Legendre, 2007). The joint effect can be positive, when the explanation is shared (i.e., the variation can be explained by any of the two factors independently), or negative, when there is an additive effect (i.e., the variation can be explained only by the inclusion of both factors).

2.6. Regionalization

The pure effect of each factor can be plotted on a map, showing where and when the pure effect of a factor is higher than the effect of the remaining factors (Acevedo and Real, 2011). However, in this case, the number of counties in the NWHSP for which there were continuous data every year led to the aggregation of the effects at the province level, indicating for each province the proportion of county-year units in the fourth quartile of positive residuals (i.e., a very high pure effect against the remaining factors) with respect to the total number of county-year records in the province.

All procedures were carried out using R version 3.6.2 (R Core Team, 2013), the "tidyverse" (Wickham, 2017) and the "INLA" R packages (Rue et al., 2009).

3. Results

3.1. Descriptive analysis

The information available for cattle included 490 counties sampled throughout the study period (2014–2019), providing 2940 county-year records. However, after excluding those records in which fewer than 10 wild boar were tested for TB, which included the records appertaining to all the islands, information on disease status in cattle and wild boar was available for 315 counties. Most of these counties did not provide data for the whole study period, signifying that 906 county-year records were available for analysis.

The prevalence, incidence, and number of counties that maintain positive TB results in cattle herds for each year are shown in Table 2. The spatial representation of the data shows a higher prevalence and incidence in cattle herds in the southwestern quadrant of Spain (see Fig. 2). The annual spatial representation of TB prevalence and incidence in cattle can be found in Appendix 1. Maintenance and persistence are likewise higher in these regions, but also in central-western Spain and in the central part of the north coast of Spain, for which it is necessary to highlight regions with a low prevalence and incidence but which have difficulties in achieving the OTF status. No significant differences were found between the mean values for the overall counties sampled and the counties used for the modeling in any of the years (unpaired two-samples t-test; p-value > 0.05).

With regard to TB prevalence in wild boar, the temporal trend models revealed a significant increase in the six-year study period, with two stages of increase, from 2014 to 2015 and from 2018 to 2019 (see Appendix 4). With reference to the spatial distribution of the prevalence, we observed a spatial variation of the counties sampled each year, although the prevalence is clearly higher in central-western Spain (Figs. 3 and 4).

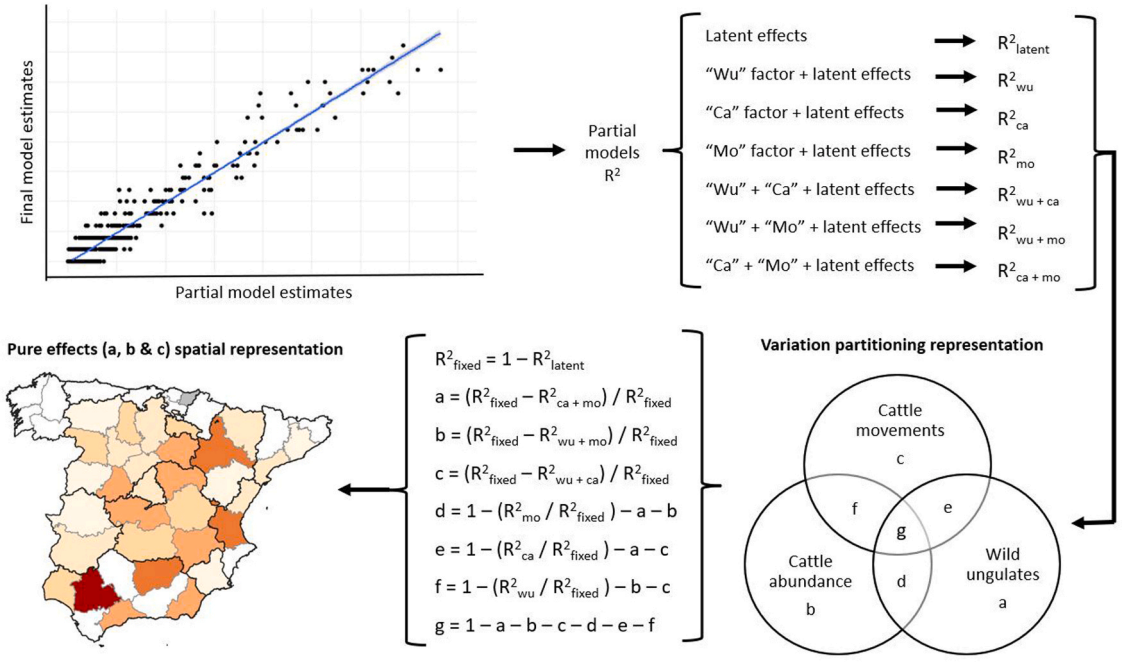


Fig. 1. Fixed factor variation partitioning procedures (based on Heikkinen et al., 2005). "wu", "ca" and "mo" refer to the wild ungulate, cattle abundance and cattle movement factors, respectively. The upper side describes the R^2 estimation process, in which predicted values of the final model are compared with the predicted values of partial models (excluding one and two factors from the complete model), and the R^2 of the linear regression between both predictions is calculated. The lower side describes how the weight of each factor is estimated, considering that higher values of R^2 between the final model and the partial model are related to a higher weight in the final model prediction owing to the factors included in the partial model, given the greater similarity between both predictions. The spatial representation of the pure effects (a, b & c, which is the explanation of the final model that can be provided only by one factor) makes it possible to identify the regions in which each factor is driving the model predictions.

Table 2
Animal tuberculosis positive-tested herds (Pos), new positive-tested herds (New pos), number of herds tested (Herds), annual prevalence (Prev) and incidence (Inc), both in %, number of counties that maintain TB positive results (Mt), number of counties sampled (NC) and percentage of counties sampled that maintain TB positive results (%) for Spain, and for the counties used for the modeling. Graphs of the temporal patterns are shown in Figure S.1. in Appendix 1.

Year	Overall Spanish counties sampled								Counties used for modeling							
	Pos	New pos	Herds	Prev	Inc	Mt	NC	%	Pos	New pos	Herds	Prev	Inc	Mt	NC	%
2014	1863	1146	108,748	1.71	-	-	487	-	413	204	22,725	1.82	-	-	113	-
2015	3070	2204	109,063	2.81	2.06	174	488	35.7	1476	1030	38,545	3.83	2.70	80	161	49.7
2016	3044	1689	106,304	2.86	1.64	189	488	38.7	1809	1000	45,802	3.95	2.26	92	182	50.5
2017	2461	1223	106,778	2.30	1.18	163	488	33.4	1132	545	30,935	3.66	1.87	54	117	46.2
2018	2384	1400	104,069	2.29	1.38	139	490	28.4	1660	1043	48,566	3.42	2.20	74	189	39.2
2019	1874	907	98,643	1.90	0.94	129	490	26.3	1214	560	42,405	2.86	1.37	54	144	37.5

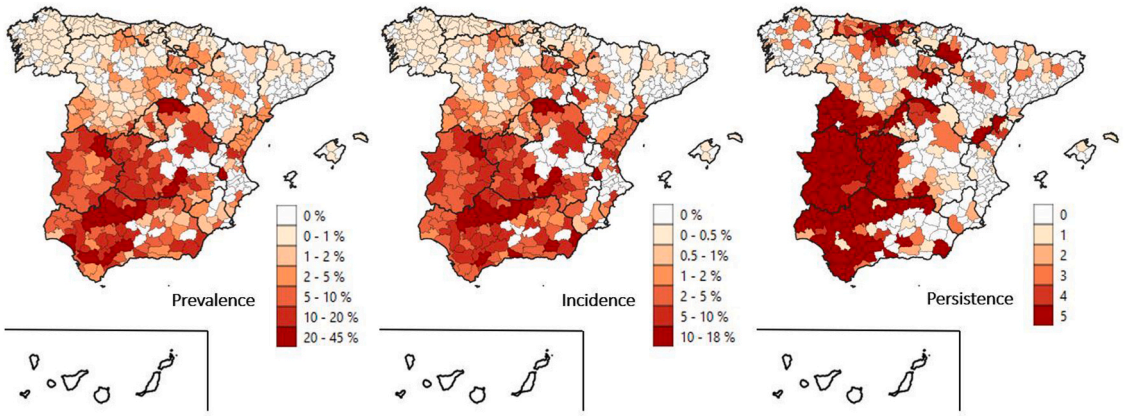


Fig. 2. Spatial distribution of prevalence (% positive herds), incidence (% new positive herds) and persistence (sum of consecutive years in which the disease is maintained) of animal tuberculosis in cattle herds during the studied period in Spain. Prevalence and incidence are shown as the mean value of the years studied. The Canary Islands are represented in the box under the map.

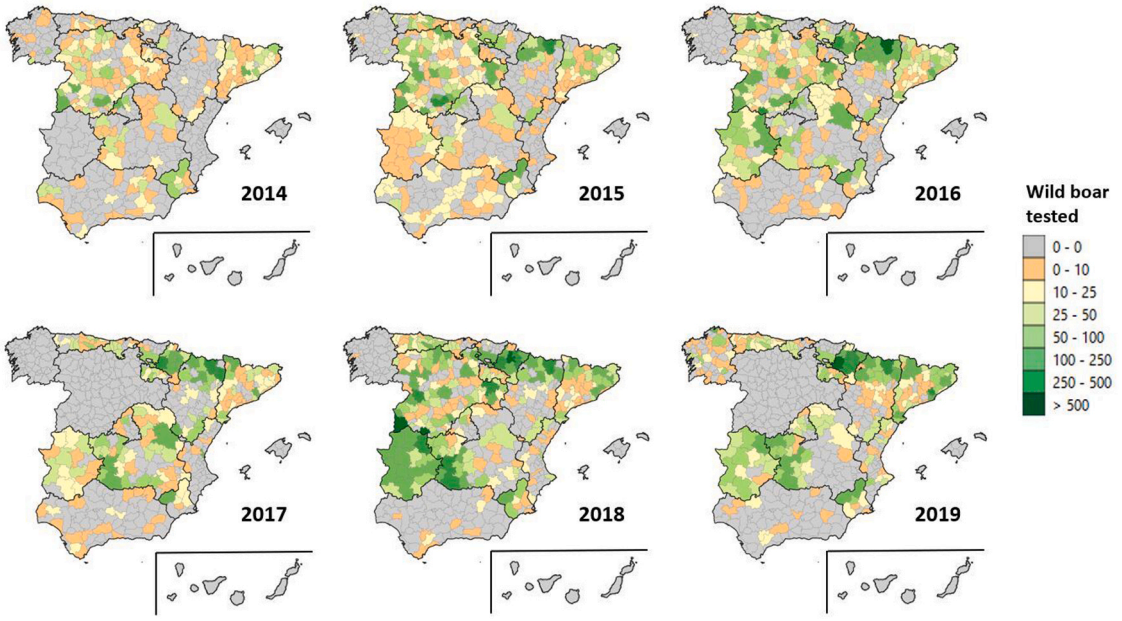


Fig. 3. Spatial distribution of the number of wild boar tested in the National Wildlife Health Surveillance framework during the period 2014 – 2019. Counties with less than 10 wild boar tested (grey and orange) were excluded from subsequent modeling. The Canary Islands are represented in the box under the map.

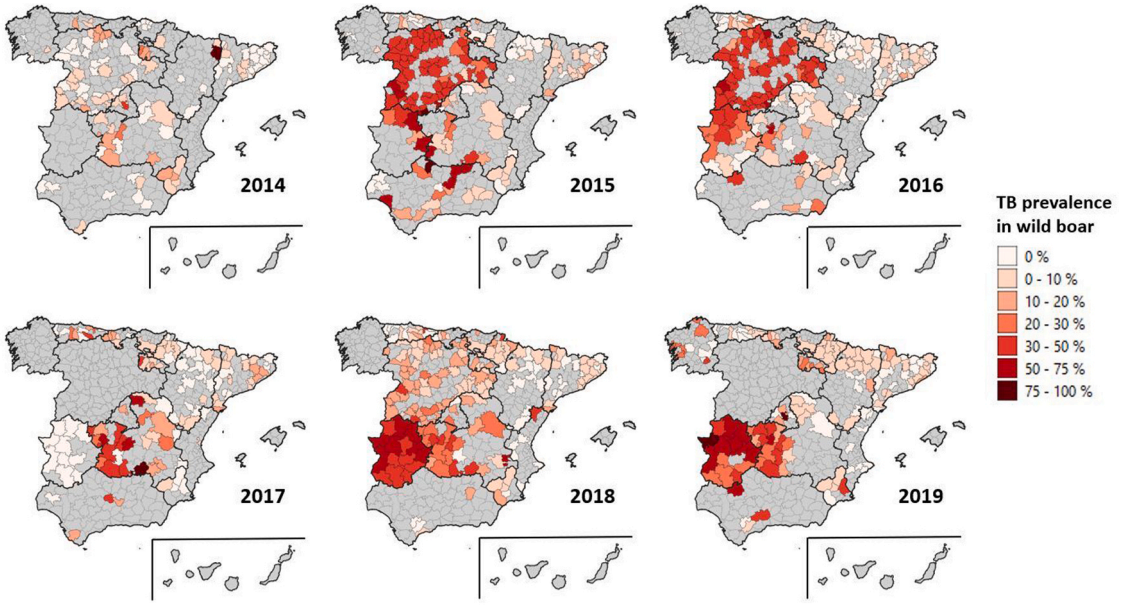


Fig. 4. Spatial distribution of TB prevalence in wild boar, as the percentage of positives with respect to the total number of samples, in counties with at least 10 samples tested in the National Wildlife Health Surveillance during the period 2014 – 2019. Counties with less than 10 samples are represented in grey. The Canary Islands are represented in the box under the map.

Table 3
Number of wild boar that tested positive to TB (Pos), number of wild boar tested (Nwb), prevalence in % (Prev), and number of counties sampled (NC) for the whole of Spain and for the counties used in our models.

Year	All Spanish counties sampled				Counties used for modeling			
	Pos	Nwb	Prev	NC	Pos	Nwb	Prev	NC
2014	196	3839	5.11	225	178	3427	5.19	113
2015	1819	8192	22.2	278	1738	7731	22.4	161
2016	1817	10678	17.01	262	1739	10369	16.77	182
2017	643	6652	9.67	199	597	6371	9.37	117
2018	2857	17060	16.75	274	2827	16723	16.9	189
2019	1334	10673	12.5	226	1326	10394	12.76	144

3.2. Risk factor modeling

Table 4 shows the fixed and latent effects included in the final models ranked in the order in which they were selected in the step-forward procedure. The DIC of the fixed effect selection procedure are shown in Table S.2 in Appendix 3. The prevalence and incidence models included the same fixed effects, but different latent effects. While both included space-time interaction and spatial autocorrelation, the temporal effect was included only in the prevalence model. The unstructured spatial effect was not included in any of the models. Maintenance and persistence also included similar fixed effects. However, while the maintenance model included space-time interaction, temporal effect, and spatial autocorrelation, it was possible to include only the last in the persistence model.

High correlation (correlation coefficient $> |0.6|$) was detected between HR and LR movements and EHR and ELR movements, respectively. The retention of one of them in the model, therefore, led to the exclusion of the other. LR and ELR also had a high correlation with the abundance of beef herds and dairy herds.

The 95% credible intervals of the posterior probability distribution for the temporal latent effect included 0 in all the years, indicating no significant changes among years as regards the risk of being infected or maintaining the disease for the whole area studied. Nonetheless, there were significant variations in the space-time interaction latent effect at the county-year level, but for an only few sample units. This effect was not evidenced in the maintenance model.

The one-way ANOVA revealed that there was no statistically significant difference among the residual distribution of the models according to the diagnostic techniques employed for wild boar testing, neither for the prevalence (p-value = 0.23) nor for the incidence (p-value = 0.18).

3.3. Variation partitioning

The wild ungulates factor explained most of the variation in the incidence model (see Fig. 5), while cattle movement was the most explicative factor in the other models, although with differences: in the case of prevalence, the wild ungulates factor also explained a substantial part of the variance, while in that of maintenance and persistence, the remaining factors had considerably less weight.

3.4. Regionalization

Upon aggregating the results at province level, data was unavailable only for the Spanish Islands (the Canary Islands and the Balearic Islands) and one province within the Iberian Peninsula. The remaining provinces had 18.2 ± 14.7 county-year records each. The results of the regionalization procedure are plotted in Fig. 6.

4. Discussion

Research focused on the role played by wild animals in hindering TB eradication is essential as regards focusing efforts on achieving the OTF status. This study has, for the first time, employed nationwide information regarding TB prevalence in wild boar in a risk factor analysis of TB dynamics in cattle. The resulting models showed that the main driver of the disease dynamics is cattle movements, which was the most explicative factor for the prevalence, maintenance, and persistence models. However, the variation partitioning revealed the relevance of wildlife for all parameters, and this was the most explanatory factor for incidence, thus suggesting that wild ungulates in general, and wild boar in particular, should also be considered in TB management.

In Spain, information on wildlife has, to date, been available only partially through the implementation of wildlife abundance data in national scale modeling (Garcia-Saenz et al., 2014), or the data collected by employing epidemiological questionnaires (Guta et al., 2014). The NWHSP now provides nationwide information regarding the wildlife disease status. However, given the technical and financial difficulties associated with monitoring diseases in wildlife (Morner et al., 2002; Stephen et al., 2019), the data provided by the NWHSP have some limitations, associated mainly with the number of samples, their spatio-temporal distribution, and the lack of homogeneity in the diagnostic techniques employed, that should be considered and handled when carrying out statistical analyses. In this study we show a procedure with which to manage the limitations commonly present in wildlife disease data (Boadella et al., 2011b; Jamin and Riviere, 2020).

Firstly, the representativeness of the units containing information regarding wildlife disease status and concerning all the units at a national scale should be verified year by year in order to avoid significant differences. With regard to units containing disease information for wildlife, records with a small sample size (e.g., $n < 10$) should be excluded from the modeling so as to reduce bias. Avoiding small sample

Table 4
Fixed and latent effects included in the models ranked in the order in which they were included, with their coefficients, standard deviations (SD) and 95% credible intervals. Coefficients are odds ratios for the fixed effects and marginal variance (σ^2) for the latent effects.

Prevalence model					Incidence model				
Variable	Coefficient	SD	Credible 2.5%	Credible 97.5%	Variable	Coefficient	SD	Credible 2.5%	Credible 97.5%
LR movements	**0.57**	**0.06**	**0.45**	**0.70**	**LR movements**	**0.58**	**0.07**	**0.46**	**0.73**
Red deer ab.	1.01	0.13	0.77	1.29	Red deer ab.	1.12	0.13	0.88	1.41
Bullfight herds	1.07	0.06	0.95	1.21	Bullfight herds	1.07	0.06	0.95	1.19
TB in wild boar	1.02	0.04	0.95	1.11	**TB in wild boar**	**1.11**	**0.06**	**1.00**	**1.24**
Wild boar ab.	1.31	0.18	0.99	1.69	Wild boar ab.	1.27	0.18	0.96	1.65
EHR movements	1.03	0.06	0.92	1.15	EHR movements	1.04	0.06	0.93	1.17
Space-time int. (σ^2)	0.41	0.06	0.31	0.54	Space-time int. (σ^2)	0.50	0.08	0.36	0.69
Spatial autocorr. (σ^2)	4.97	1.45	2.79	8.44	Spatial autocorr. (σ^2)	4.69	2.66	1.60	11.74
Temporal effect (σ^2)	0.06	0.03	0.02	0.14					
Maintenance model					**Persistence model**				
Variable	Coefficient	SD	Credible 2.5%	Credible 97.5%	Variable	Coefficient	SD	Credible 2.5%	Credible 97.5%
HR movements	**6.64**	**1.83**	**3.96**	**11.01**	**HR movements**	**4.72**	**1.03**	**3.11**	**7.13**
Beef herds	**1.91**	**0.48**	**1.16**	**3.06**	**Bullfight herds**	**3.56**	**1.15**	**1.91**	**6.36**
TB in wild boar	**1.82**	**0.43**	**1.13**	**2.79**	**Beef herds**	**1.74**	**0.31**	**1.21**	**2.45**
Bullfight herds	**3.49**	**1.46**	**1.64**	**7.27**	**TB in wild boar**	**1.73**	**0.34**	**1.17**	**2.50**
					Red deer ab.	1.30	0.25	0.86	1.86
Space-time int (σ^2)	0.16	0.56	0.00	1.01	Spatial autocorr. (σ^2)	3.34	0.99	1.83	5.70
Spatial autocorr. (σ^2)	5.59	1.90	2.75	10.13					
Temporal effect (σ^2)	0.92	1.65	0.02	4.65					

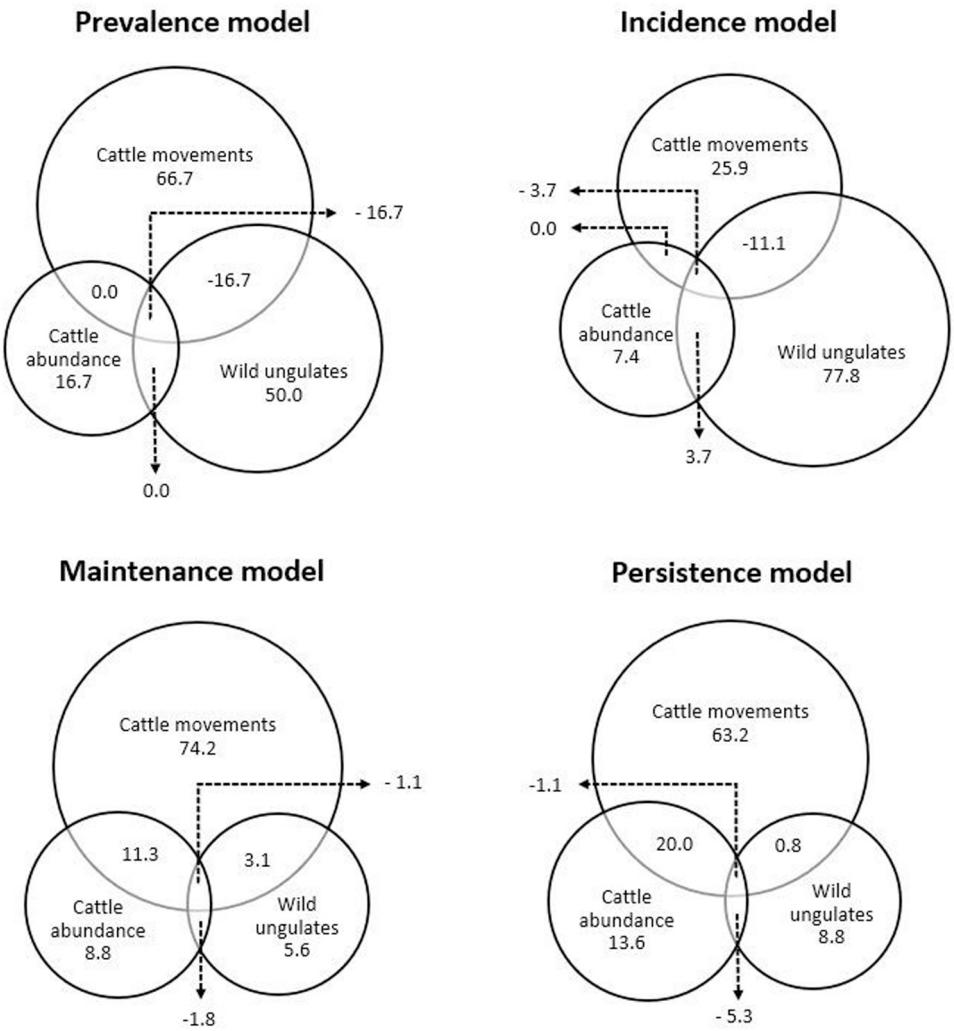


Fig. 5. Venn diagrams representing the partition of the variation (%) among each factor for the four models. The values under the name of the factor represent the pure effect of each factor, and the numbers in the overlapped area represent the overlaid effect (shared, when positive, or additive, when negative).

sizes prevalence values may be inaccurate, but are unlikely to be biased. This lack of accuracy can be managed through the introduction of highly informative spatial, temporal, and spatio-temporal latent effects, which embrace much of the unexplained variation when a fixed effect is not explicative (Blangiardo and Cameletti, 2015). This results in a fitted model in which each fixed effect contributes to the best of its ability, and the latent effects complete the explanation (Abellan et al., 2008). This may, therefore, lead to the attainment of imprecise relations, but has the advantage of avoiding biased relations. Latent effects similarly embrace the effect of the spatio-temporal variation in the diagnostic techniques. A comparison of the model residuals by means of a diagnostic technique can verify this assumption if no significant differences are found, thus preventing a possible bias in the model. It is essential to highlight that the inclusion of highly informative latent effects, despite their utility as regards managing the data limitations, leads to a higher model complexity, thus hindering interpretation and bounding its extrapolation.

Another limitation of this kind of data is that, despite allowing the identification of relations, it hinders the determination of its directionality. Four TB parameters were, therefore, modeled (prevalence, incidence, maintenance, and persistence) with the objective of exploring the role played by the predictors in the TB dynamics in the cattle studied herein in greater depth, and the variance explained was partitioned by the factors included in the modeling, thus making it possible to hypothesize about the directionality and weight of the relations. Although modeling results in generalized relations for the whole country, the spatial decomposition of the pure effect of the factors included in the modeling using the regionalization process enables the identification of where each factor is highly related to the parameters modeled.

As a result of the variation partitioning analysis of this modeling, cattle movements were disclosed as the most relevant predictor in three of the four models, with the only exception of that concerning incidence (in which it still made a relevant contribution to total variation of risk). Livestock movements between herds has previously been associated with TB risk worldwide (Palisson et al., 2016; Avila et al., 2018; Kao et al., 2018; Fielding et al., 2020), and also in Spain (Garcia-Saenz et al., 2014; Pozo et al., 2019). The results also revealed the importance of disease status in the origin county, with high-risk movements being positively related to the disease parameters, while the low-risk movements were negatively related. In this respect, any animals that are moved in Spain are legally required to be subjected pre or post-movement to an IDT in order to avoid movements of infected animals (Anon, 2021). However, the sensitivity and specificity of the IDT enable movements of infected animals, thus posing the risk of an external source of infection (Bezos et al., 2014; Rodriguez-Campos et al., 2014; Pereira et al., 2020).

Bullfight and beef cattle abundances were also retained in the models. These breeds are commonly extensively managed, with increased contact with non-bovine hosts, thus posing a higher risk for TB infection than dairy herds (Martinez-Lopez et al., 2014; Gortazar et al.,

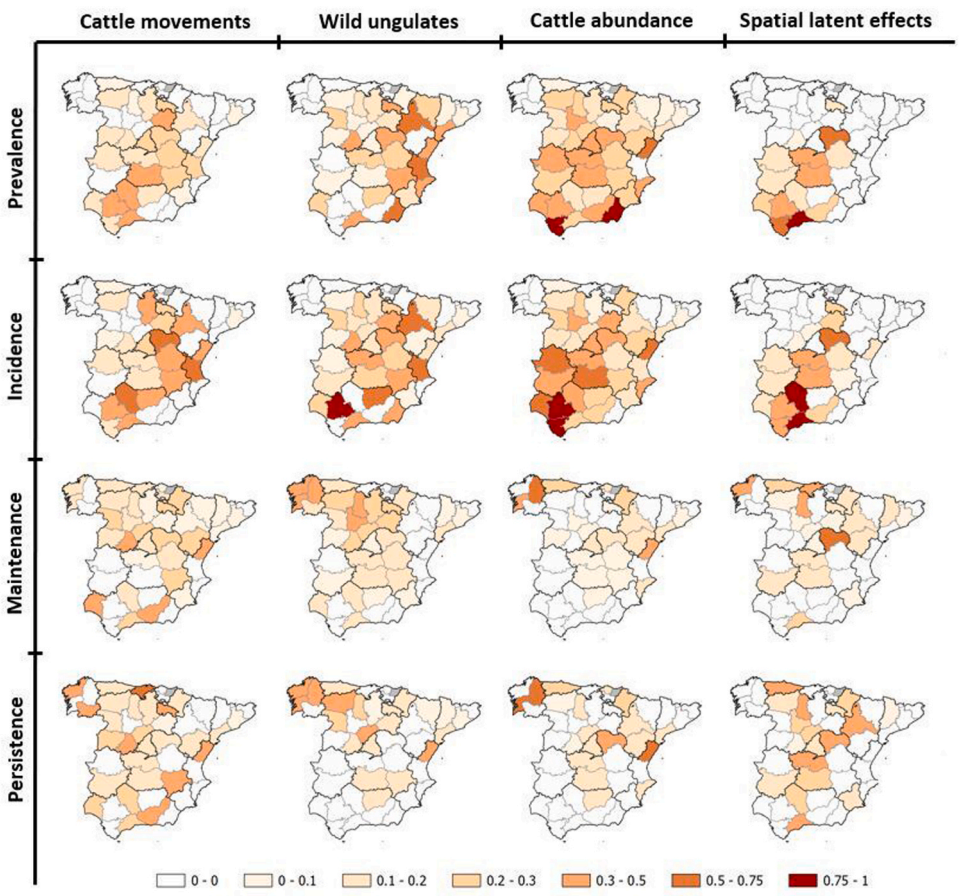


Fig. 6. Proportion of county-year records in the fourth quartile of pure effect of each factor against the others with respect to the total number of county-year records in the province. The Canary and Balearic Islands are not represented on the maps since they were excluded from the modeling.

2017; Pozo et al., 2019). Although cattle are frequently tested, the incidence of direct and indirect interactions with wildlife in extensive management systems makes it more likely that livestock will be exposed to possible infection (Barasona et al., 2017; Triguero-Ocaña et al., 2020). In this respect, improving the biosecurity of farms by implementing risk mitigation measures at the wildlife-livestock interface would be useful as regards reducing interspecific interactions (Barasona et al., 2013; Martinez-Guijosa et al., 2021). Bullfight herds are more involved and weighted in the models, and have already been linked with TB in Spain in previous studies (Allepuz et al., 2011; Rodriguez-Prieto et al., 2012; Garcia-Saenz et al., 2014). These animals are commonly managed in vast areas that are difficult to access, and their vigor and behavior make them difficult to handle, thus making TB testing more challenging (Rodriguez-Prieto et al., 2012). Furthermore, any male cattle in bullfight herds that are over 24 months of age are excluded from testing. Nonetheless, great efforts have been made to decrease the detection problems derived from extensive management, handling and testing, with the implementation of the IFN-γ, the organization of compulsory training courses for veterinarians, the reinforcement of controls in bullfight herds, and the attempt to achieve better communication among the stakeholders (Garcia-Saenz et al., 2014; Ciaravino et al., 2017; Anon, 2021).

With regard to wild ungulates, information concerning the relative abundance of wild boar and red deer were included in the modeling since both species have been demonstrated to act as TB reservoirs in Spain (Vicente et al., 2006; Vicente et al., 2007; Naranjo et al., 2008). Data regarding wild host abundance or proxies of abundance, such as hunting statistics, were sometimes used instead of disease status (Garcia-Saenz et al., 2014; Martinez-Lopez et al., 2014), since they have been shown to be related to TB prevalence on a local and regional scale (Vicente et al., 2007; Boadella et al., 2012; Vicente et al., 2013). However, on a national scale, some areas with high wild boar abundance do not have high TB prevalence, and vice versa (Gortazar et al., 2017; Varela-Castro et al., 2020). In our models, TB prevalence in wild boar and wild boar relative abundance were retained together, thus demonstrating that the disease status of the wild hosts has an additional explanatory capacity beyond the population abundance data when analyzing the disease dynamics in cattle herds on a national scale.

However, the relation found between wildlife and TB dynamics in cattle lacks directionality. Transmission both from wildlife to cattle and from cattle to wildlife have been observed in Whole Genome Sequence studies (Glaser et al., 2016; Crispell et al., 2019; Salvador et al., 2019; Crispell et al., 2020; Rossi et al., 2021), with wildlife being able to change its role under different conditions (Nugent, 2011; VerCauteren et al., 2018; Crispell et al., 2020). In our case, the variation partitioning procedures used in the four parameters analyzed may help reveal which factors hinder disease eradication in different scenarios: wild ungulates had a great explanatory capacity for prevalence, and especially for incidence, but not for maintenance and persistence. Wild ungulates have been documented to come into contact with cattle (Barasona et al., 2013; Wilber et al., 2019; Triguero-Ocaña et al., 2020), and are, therefore, a possible route of entry of the pathogen into herds that are uninfected. However, in already infected herds, other factors can operate to a greater extent in disease prevalence and maintenance, and cattle movements are more explanatory in these cases, along with other possible factors not disclosed in our models (although they can be partially gathered by the latent effects), such as the presence of fenced estates, the percentage of extensive cattle farms, the use of communal pastures, or the presence of small ruminants (Gortazar et al., 2017). It is, therefore, possible that once the disease is established in cattle, the

relationship between wild boar TB prevalence and TB in cattle is owing to cattle to wild boar transmission. These results highlight the role played by cattle movements in disease management, and the importance of considering wildlife when analyzing breakdowns.

A geographical representation of the pure effect of factors has previously been used in ecology (Acevedo and Real, 2011), but its use has not been extended to spatial epidemiology. Since pure effects provide a robust measure of the contribution of each predictor (Heikkinen et al., 2005; Murray and Conner, 2009), their representation would be of interest as regards identifying the risk factors driving the presence and dynamics of a pathogen (Peterson, 2008). The procedure employed allows the pure effect of a factor to show those cases in which the expected value for a given disease parameter analyzed is higher than that which would be expected for the remaining factors and the overlaid effects among them. The collinear variation and synergic effects make it difficult to discriminate the isolated influence of the factors (Hortal et al., 2008), and spatial autocorrelation in predictors can affect the predictor–response relationship (Harisena et al., 2021). A low value for a pure effect of a given factor does not, therefore, necessarily mean a low explanatory capacity, since a substantial part of the explanation could remain in overlaid or latent effects. However, high values in pure effects do show when a given factor is more explanatory than the others (Legendre, 2007), thus making it possible to know which factor is mainly related to the pathogen dynamics in a determined area. Measures aimed at achieving the OTF status can consequently be focused on the main factors determining the disease dynamics for each area, based on the regionalization criterion. Regionalization is, therefore, a very useful tool with which to identify risk factors by determining the presence and dynamics of a pathogen in certain areas, and provides a new means of analysis with the aim of managing the disease that can be improved as the coverage of the NWHSP increases.

Funding

CH is supported by a predoctoral grant funded by European Social Fund (ESF) and Castilla-La Mancha regional government (JCCM; ref: SUPLY/19/180501/000487).

CRediT author statement

CH: Conceptualization, Methodology, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **JV:** Conceptualization, Investigation, Data curation, Writing – review & editing. **CG:** Investigation, Writing – review & editing. **PA:** Conceptualization, Methodology, Validation, Investigation, Writing – original draft, Writing – review & editing, Project administration.

Declaration of Competing Interests

The authors declare that they have no competing interests.

Acknowledgments

We would like to thank the Spanish Ministry of Agriculture, Food and Fisheries (MAPA; Ministerio de Agricultura, Pesca y Alimentación) for its continuous support by means of different contracts and data provision, and project PID2019-111699RB-I00 (Proyectos de I+D+i Retos Investigación tipo B). We are particularly grateful to Germán Cáceres Garrido and José Luís Sáez Llorente from the Spanish Ministry of Agriculture, Food and Fisheries. We also thank Sally Newton for English proofreading.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.prevetmed.2023.106049.

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